Course Review

Econ 502: Advanced Microeconomics

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What We Covered

- **Part I - Competitive Markets** (Lectures 1–5): consumer theory, production, partial equilibrium, general equilibrium, welfare theorems
- **Part II - Market Power** (Lectures 6–7): monopoly, price discrimination, oligopoly (Cournot, Bertrand, Stackelberg)
- **Part III - Market Failures** (Lectures 8–9): adverse selection, externalities, public goods
- **Part IV - Game Theory** (Lecture 10): Nash equilibrium, mixed strategies, sequential games, repeated games

What to Expect on the Exam

- **Final exam content:** Lecture 4 + Parts II–IV (lectures 6–10)
- **Comprehensive exam content:** all lectures
- **Format:** 4-5 problems, similar to problem sets and practice problems. Closed book, 2 hours.

Part I: Competitive Markets

Consumer Theory (Lec 1-2)

- **Preferences and utility:** axioms (completeness, transitivity, monotonicity, convexity); MRS = ratio of marginal utilities = slope of indifference curve
- **Utility maximization (primal):** $\max U(x, y)$ s.t. $p_x x + p_y y = I$. Tangency: $MRS = p_x / p_y$. Yields **Marshallian demand** $x^M(p, I)$ and **indirect utility** $V(p, I)$
- **Expenditure minimization (dual):** $\min p_x x + p_y y$ s.t. $U(x, y) = \bar{u}$. Yields **Hicksian demand** $x^h(p, \bar{u})$ and **expenditure function** $e(p, \bar{u})$
- **Duality:** V and e are inverses in income/utility
- **Slutsky decomposition:** $\frac{\partial x^M}{\partial p} = \underbrace{\frac{\partial x^h}{\partial p}}_{<0 \text{ subst.}} - x \cdot \underbrace{\frac{\partial x^M}{\partial I}}_{\text{income}}$

Welfare Measures

For a price change $p_0 \longrightarrow p_1$:

- **Compensating Variation:** $CV = e(p_1, u_0) - e(p_0, u_0)$ - extra income at *new* prices to keep old utility
- **Equivalent Variation:** $EV = e(p_1, u_1) - e(p_0, u_1)$ - income change at *old* prices that replicates new utility
- **Consumer surplus:** area below Marshallian demand above price; lies between CV and EV for normal goods

When are CV and CS equal?

Things to Remember

Cobb-Douglas shortcut: $U(x, y) = x^{\alpha}y^{\beta}$:

$$x^M = \frac{\alpha}{\alpha + \beta} \cdot \frac{I}{p_x}, \quad y^M = \frac{\beta}{\alpha + \beta} \cdot \frac{I}{p_y}$$

With $\alpha + \beta = 1$,

$$x^M = \frac{\alpha I}{p_x}, \quad y^M = \frac{\beta I}{p_y}$$

Production and Cost (Lec 3)

- **Production function** $Q = f(K, L)$: marginal product, MRTS, isoquants
- **Returns to scale:** if $f(\lambda K, \lambda L) = \lambda^k f(K, L)$, RTS = k (decreasing if < 1, constant if $= 1$, increasing if > 1)
- **Cost minimization:** tangency $MRTS = w/r$ + isoquant constraint
- **Cost function:** $C(Q, w, r)$. Average cost AC, marginal cost MC. Note: MC crosses AC at minimum

Note: RTS is about scaling *all* inputs; DRS in production ≠ DRS in $C(Q)$ (cost function curvature is opposite).

Market Equilibrium and Tax Incidence (Lec 4)

- **Equilibrium:** $Q^D(P) = Q^S(P)$. Surplus = areas under demand and above supply
- **Per-unit tax** t on producers: solve $Q^D(P^D) = Q^S(P^D - t)$
- **Incidence formula:** consumer share = $\dfrac{\varepsilon_S}{|\varepsilon_D| + \varepsilon_S}$, producer share $= \dfrac{|\varepsilon_D|}{|\varepsilon_D| + \varepsilon_S}$
- **DWL** = $\frac{1}{2}(\Delta Q)(t)$; equivalently $\frac{1}{2} \cdot \dfrac{|\varepsilon_D|\varepsilon_S}{|\varepsilon_D| + \varepsilon_S} \cdot \dfrac{t^2}{P^*} \cdot Q^*$

Key insight: **statutory** incidence (who writes the check) ≠ **economic** incidence (who bears the burden). The latter depends entirely on relative elasticities.

General Equilibrium and Welfare Theorems (Lec 5)

- **Edgeworth box:** allocations where indifference curves are tangent (= equal MRS) are Pareto efficient - the **contract curve**
- **First Welfare Theorem:** any competitive (Walrasian) equilibrium with locally non-satiated preferences is Pareto efficient
- **Second Welfare Theorem:** any Pareto efficient allocation can be supported as a competitive equilibrium after appropriate **lump-sum transfers** of endowments (requires convex preferences/technology)

Slogan: *Markets handle efficiency; government handles equity*. Caveat: real lump-sum transfers are hard (require knowing types), and many real-world settings violate FWT assumptions (externalities, public goods, market power, asymmetric info - Parts II–IV).

Part II: Market Power

Monopoly (Lec 6)

- **Profit max:** $MR = MC$
- **MR for linear demand** $P = a - bQ$: $MR = a - 2bQ$ (twice the slope, same intercept)
- **Lerner index:** $L = (P - MC)/P = 1/|\varepsilon_D|$ (inverse-elasticity rule)
- **DWL** of monopoly: triangle between demand and MC over output not produced
- Monopolist always sets P on the **elastic** portion of demand ($|\varepsilon_D| > 1$)

Price discrimination:

- 1st degree: charge each buyer their WTP (full surplus extraction)
- 2nd degree: self-selection menus (e.g., quantity discounts)
- 3rd degree: segment by observable type, charge different prices; $L_i = 1/|\varepsilon_i|$ in each segment

Oligopoly (Lec 7) - Cournot

Two firms, inverse demand $P = a - bQ, Q = q_1 + q_2$, equal MC c:

- **Best response:** $q_i^*(q_{-i}) = \dfrac{a - c - q_{-i}}{2b}$
- **Symmetric NE:** $q_i^* = \dfrac{a - c}{3b}, Q^* = \dfrac{2(a - c)}{3b}$
- **Quantities are strategic substitutes** (BR slopes downward)
- With n firms: $Q^* = \dfrac{n(a - c)}{(n + 1)b} \to \dfrac{a - c}{b}$ as $n \to \infty$ (competitive limit)

Oligopoly (Lec 7) - Bertrand and Stackelberg

Bertrand (homogeneous): $p_1 = p_2 = c$ (Bertrand paradox - two firms enough for competitive outcome)

Differentiated Bertrand: demand $q_i = a - bp_i + dp_{-i}$. Prices are **strategic complements** (BR slopes up). $p^* > c$ - paradox resolved by differentiation.

Stackelberg: leader moves first. Leader's problem substitutes follower's BR. Result: leader produces *more* than Cournot, follower less. Leader profit *higher*, follower *lower*. ` `Top dog'' commitment.

Note: the leader gains from commitment because quantities are strategic substitutes. With strategic complements (prices), being the price leader can *hurt* you.

Part III: Market Failures

Adverse Selection (Lec 8)

- **Lemons problem (Akerlof):** when sellers know quality but buyers don't, the average quality of cars on the market is below the unconditional average $\Rightarrow$ market unravels
- **Insurance:** types differ in risk p_i. Risk-averse u concave $\Rightarrow$ everyone willing to pay above the actuarially fair premium $p_i L$ (the gap is the **risk premium**)
- **Certainty equivalent:** CE_i solves $u(CE_i) = E[u(z)]$. Max premium $P_i^{\max} = w - CE_i$. High-risk types have higher $P_i^{\max}$.
- **Pooling equilibrium:** if insurer can't price discriminate, must set P based on average risk. If $P > P_L^{\max}$, low-risk types drop out and market unravels.

Externalities (Lec 9)

- **Externality:** private MC $\neq$ social MC. Unregulated market over-produces (negative ext.) or under-produces (positive)
- **Coase theorem:** with well-defined property rights and zero transaction costs, private bargaining yields efficiency *regardless of who holds the rights* (only distribution differs)
- **Pigouvian tax:** set $t =$ marginal damage; firms internalize externality
- **Cap and trade:** issue permits = efficient pollution; market trading achieves efficient allocation regardless of initial allocation (Coase logic)

Tax fixes **price**, lets quantity adjust. Cap fixes **quantity**, lets price adjust. Equivalent under certainty.

Public Goods (Lec 9)

Non-rival + non-excludable goods.

- **Voluntary contribution:** each agent sets their own MRS_i = price ($= 1$ when public good costs $1 per unit). Severe under-provision
- **Samuelson condition (efficient):** $\sum_i MRS_i = 1$ - sum of willingness-to-pay equals marginal cost
- **Free-rider problem:** each agent wants to benefit from others' contributions without paying. Leads to under-provision of the public good.

Part IV: Game Theory (Lec 10)

Normal-Form Games

- **Strict dominance:** strategy s strictly dominates s' if $u(s, s_{-i}) > u(s', s_{-i})$ for all s_{-i}
- **Nash equilibrium:** profile $(s_1^*, \ldots, s_n^*)$ such that no one wants to unilaterally deviate. Each plays best response to others
- **Mixed strategies:** when no pure NE exists (e.g., Matching Pennies). Each player **indifferent** in equilibrium between strategies in their support

Note: a Nash strategy doesn't have to be dominant - only a best response *to the equilibrium profile*. Dominance is much stronger.

Sequential Games

- **Extensive form (game tree):** decision nodes, branches, payoffs at terminals
- **Backward induction:** solve from the end. At each decision node, the player picks her best action given anticipated future play
- **Subgame Perfect Equilibrium (SPE):** Nash in *every subgame*. Rules out non-credible threats
- **Commitment:** moving first can be valuable when actions are strategic substitutes (Stackelberg top dog''); hurtful when complements (puppy dog”)

Entry deterrence: sunk capacity makes the threat of post-entry aggression credible. Words alone do not.